

a)

$$\frac{D}{Dt}(K + G + E) = \dot{Q} + \dot{W}$$

$$\frac{D}{Dt}K + \frac{D}{Dt}G + \frac{D}{Dt}E$$

$$K = \int_V \frac{1}{2} \rho v_i v_i dV$$

$$\frac{D}{Dt} \int_V A dV = \int_V \left(\frac{DA}{Dt} + A \frac{\partial v_j}{\partial x_j} \right) dV.$$

$$\frac{D}{Dt}K = \int_V \left[\frac{D}{Dt} \left(\frac{1}{2} \rho v_i v_i \right) + \frac{1}{2} \rho v_i v_i \cdot \frac{\partial v_j}{\partial x_j} \right] dV$$

$$\int_V \frac{D}{Dt} \left(\rho \frac{1}{2} v_i v_i \right) dV = \int_V \rho \cdot \frac{D}{Dt} \left(\frac{1}{2} v_i v_i \right) dV = \int_V \rho \cdot v_i \frac{D(v_i)}{Dt} dV$$

Ec. Continuidad

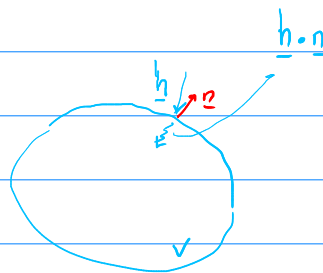
$$\frac{1}{2} \left(\frac{D(v_i)}{Dt} v_i + v_i \frac{D(v_i)}{Dt} \right)$$

$$\frac{D}{Dt}G = \frac{D}{Dt} \int_V (\rho \phi) dV = \int_V \rho \cdot \frac{D\phi}{Dt} dV$$

$$\frac{D}{Dt}E = \frac{D}{Dt} \int_V \rho \epsilon dV = \int_V \rho \cdot \frac{D\epsilon}{Dt} dV$$

Ten. de flujo

$$\dot{Q} = - \int_S h_i n_i dS = - \int_V \frac{\partial h_i}{\partial x_i} dV$$



$$\dot{W} = \int_V F_i v_i dV + \int_S \frac{\sigma}{2} v_i dS = \int_V \left[F_i v_i + \frac{\partial (\pi_{ji} v_i)}{\partial x_j} \right] dV =$$

$$\frac{\partial \pi_{ji}}{\partial x_j} v_i + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

$\forall V \Rightarrow$

$$\rho \cdot v_i \frac{Dv_i}{Dt} + \rho \cdot \frac{D\phi}{Dt} + \rho \cdot \frac{D\epsilon}{Dt} = - \frac{\partial h_i}{\partial x_i} + F_i v_i + \frac{\partial \pi_{ji}}{\partial x_j} v_i + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

De la ec. de movimiento:

$$\rho \cdot \frac{Dv_i}{Dt} = \frac{\partial \pi_{ij}}{\partial x_j} + \rho x_i \Rightarrow \rho \cdot \frac{Dv_i}{Dt} - x_i = \frac{\partial \pi_{ij}}{\partial x_j}$$

$$= - \frac{\partial h_i}{\partial x_i} + F_i v_i + \left(\rho \cdot \frac{Dv_i}{Dt} - x_i \right) v_i + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

$$= - \frac{\partial h_i}{\partial x_i} + (F_i - x_i) \cdot v_i + \rho v_i \frac{Dv_i}{Dt} + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

$$g_i = -\rho \cdot \frac{\partial \phi}{\partial x_i}$$

$$X_i = F_i + g_i \Rightarrow F_i - X_i = -g_i = \rho \cdot \frac{\partial \phi}{\partial x_i}$$

$$\cancel{\rho \cdot v_i \frac{Dv_i}{Dt}} + \cancel{\rho \cdot \frac{D\phi}{Dt}} + \cancel{\rho \cdot \frac{D(\epsilon)}{Dt}} = -\frac{\partial h_i}{\partial x_i} + \cancel{\rho \cdot \frac{\partial \phi}{\partial x_i} v_i} + \cancel{\rho \cdot v_i \frac{\partial v_i}{\partial t}} + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

$\rho \cdot \frac{\partial \phi}{\partial t} + \rho \cdot \frac{\partial \phi}{\partial x_i} v_i$ solo depende de x

$$\rho \cdot \frac{D(\epsilon)}{Dt} = -\frac{\partial h_i}{\partial x_i} + \pi_{ji} \frac{\partial v_i}{\partial x_j}$$

Consejo que π es simétrico

$$\pi_{ji} \frac{\partial v_i}{\partial x_j} = \pi_{ji} \cdot \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) = \pi_{ji} \cdot \frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right)$$

se agrega 0

$$= \pi_{ji} \left[\underbrace{\frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right)}_{V_{ij} \text{ (simétrico)}} + \underbrace{\frac{1}{2} \left(\frac{\partial v_i}{\partial x_j} - \frac{\partial v_j}{\partial x_i} \right)}_{-R_{ij} \text{ (antisimétrico)}} \right] =$$

$$= \pi_{ji} (V_{ij} - R_{ij}) = \underbrace{\pi_{ji} V_{ij}}_{\pi_{ij} \text{ (simétrico de } \pi \text{)}} - \underbrace{\pi_{ji} R_{ij}}_{0 \text{ (contracción de tensor simétrico con un antisimétrico)}}$$

$$\pi_{ji} R_{ij} = \pi_{ji} \left[\frac{1}{2} (R_{ij} + R_{ij}) \right] = \frac{1}{2} (\underbrace{\pi_{ji} R_{ij}}_{\pi_{ij}} + \underbrace{\pi_{ji} R_{ij}}_{\pi_{ij}}) = \frac{1}{2} (\underbrace{\pi_{ji} R_{ij}}_{\pi_{ij}} - \underbrace{\pi_{ij} R_{ji}}_{\pi_{ij}}) = 0$$

(Por la antisimetría de R)

$$\rho \cdot \frac{D\epsilon}{Dt} = -\frac{\partial h_i}{\partial x_i} + \pi_{ij} V_{ij}$$

b) Ley de Fourier de transferencia de calor:

$$h_i = -k \frac{\partial T}{\partial x_i}$$

↗ conductividad

$$\underline{h} = -k \underline{\nabla} T$$

$$\epsilon = c \cdot T$$

$$\rho \cdot \frac{D(c \cdot T)}{Dt} = -\frac{\partial}{\partial x_i} \left(-k \cdot \frac{\partial T}{\partial x_i} \right) + \pi_{ij} \cdot V_{ij}$$

$$\rho \cdot \left(\frac{\partial (c \cdot T)}{\partial t} + \cancel{v_j \cdot \frac{\partial (c \cdot T)}{\partial x_j}} \right) = -\frac{\partial}{\partial x_i} \left(-k \cdot \frac{\partial T}{\partial x_i} \right) \Rightarrow \rho \cdot \frac{\partial (c \cdot T)}{\partial t} = -\frac{\partial}{\partial x_i} \left(-k \frac{\partial T}{\partial x_i} \right)$$

dentro de $\rho \cdot \frac{D}{Dt} \Rightarrow v_i = 0 \Rightarrow V_{ij} = 0$